

Exploring x Raised to x : From Derivative to Antiderivative

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Abstract

The function $f(x) = x^x$ presents a blend of exponential and power-like behavior that places it outside the realm of elementary functions. In this note, we systematically explore its higher-order derivatives using Bell polynomials and the Faà di Bruno formula, and develop a differential operator approach that offers a concise perspective on its antiderivatives. Although the results are classical in essence, the unified presentation offers pedagogical clarity and highlights structural insights into a deceptively simple function.

1. Introduction

The function x^x , while appearing elementary, is notable for its nontrivial analytical properties. Unlike monomials or pure exponentials, its base and exponent are both variable, complicating the processes of differentiation and integration. This short note aims to lay out a clean, structured derivation of its derivatives and a recursive framework for its antiderivatives. Our focus is not on novelty but rather on exposing the subtle structure that lies beneath.

Derivatives of $f(x) = x^x$

$$f(x) = x^x$$

$$f'(x) = x^x(\ln x + 1)$$

$$f''(x) = x^x(\ln x + 1)^2 + x^{x-1}$$

$$f^{(3)}(x) = x^x(\ln x + 1)^3 + 2x^{x-1}(\ln x + 1) + x^{x-1}\left(\ln x + 1 - \frac{1}{x}\right)$$

$$\begin{aligned} f^{(4)}(x) &= x^x(\ln x + 1)^4 \\ &\quad + 3x^{x-1}(\ln x + 1)^2 \\ &\quad + 2x^{x-1}\left(\ln x + 1 - \frac{1}{x}\right)(\ln x + 1) \\ &\quad + \frac{2x^{x-1}}{x} \\ &\quad + x^{x-1}\left(\ln x + 1 - \frac{1}{x}\right)^2 \\ &\quad + x^{x-1}\left(\frac{1}{x} + \frac{1}{x^2}\right) \end{aligned}$$

General Formula for the Derivatives of $f(x) = x^x$

We begin by rewriting the function in exponential form:

$$f(x) = x^x = e^{x \ln x}$$

Let us define:

$$g(x) = x \ln x$$

Then, $f(x) = e^{g(x)}$. When differentiating $f(x) = e^{g(x)}$, the derivatives can be expressed using the **Faà di Bruno formula**, which leads to a representation in terms of the **complete Bell polynomials**:

$$f^{(n)}(x) = e^{g(x)} \cdot Y_n \left(g'(x), g''(x), \dots, g^{(n)}(x) \right)$$

In our case:

$$g'(x) = \ln x + 1, \quad g''(x) = \frac{1}{x}, \quad g^{(3)}(x) = -\frac{1}{x^2}, \quad g^{(4)}(x) = \frac{2}{x^3}, \quad \dots$$

Therefore, the general formula for the n th derivative of x^x is:

$$f^{(n)}(x) = x^x \cdot Y_n \left(\ln x + 1, \frac{1}{x}, -\frac{1}{x^2}, \frac{2}{x^3}, \dots \right)$$

Here, Y_n denotes the n th complete Bell polynomial, which systematically organizes all the terms arising from applying the chain and product rules repeatedly.

This formula reveals the deep combinatorial structure behind the derivatives of x^x and explains the patterns observed in the lower-order derivatives.

Recursive Construction of Antiderivative via Derivative Structure

We are given the general form of the n th derivative of the function $f(x) = x^x$, which can be expressed using the complete Bell polynomials as follows:

$$f^{(n)}(x) = x^x \cdot Y_n \left(g'(x), g''(x), \dots, g^{(n)}(x) \right)$$

where:

$$g(x) = x \ln x, \quad g'(x) = \ln x + 1, \quad g''(x) = \frac{1}{x}, \quad g^{(3)}(x) = -\frac{1}{x^2}, \quad g^{(4)}(x) = \frac{2}{x^3}, \quad \dots$$

Our goal is to reverse engineer this expression to obtain the original function $f(x)$, i.e., to integrate the n th derivative n times.

Step 1: Assume Structure of the Integral

Suppose the $(n-1)$ st derivative can be written as:

$$f^{(n-1)}(x) = x^x \cdot A_{n-1}(x)$$

Taking the derivative:

$$f^{(n)}(x) = \frac{d}{dx} (x^x \cdot A_{n-1}(x)) = x^x (\ln x + 1) A_{n-1}(x) + x^x A'_{n-1}(x) = x^x (A'_{n-1}(x) + (\ln x + 1) A_{n-1}(x))$$

Set this equal to the known expression:

$$x^x (A'_{n-1}(x) + (\ln x + 1) A_{n-1}(x)) = x^x \cdot Y_n(x) \Rightarrow A'_{n-1}(x) + (\ln x + 1) A_{n-1}(x) = Y_n(x)$$

Step 2: Solve the Linear ODE

This is a first-order linear ODE. We solve it using an integrating factor:

$$\mu(x) = e^{\int (\ln x + 1) dx} = e^{x \ln x - x} = x^x e^{-x}$$

Multiply both sides by the integrating factor:

$$\frac{d}{dx} (A_{n-1}(x) \cdot x^x e^{-x}) = x^x e^{-x} \cdot Y_n(x) \Rightarrow A_{n-1}(x) = \frac{1}{x^x e^{-x}} \int x^x e^{-x} Y_n(x) dx$$

Therefore:

$$f^{(n-1)}(x) = x^x \cdot A_{n-1}(x) = e^x \int x^x e^{-x} Y_n(x) dx$$

Step 3: Generalizing by Recursion

We can recursively repeat this step to obtain:

$$f^{(n-2)}(x) = \int f^{(n-1)}(x) dx = \int e^x \left(\int x^x e^{-x} Y_n(x) dx \right) dx$$

Continuing this process, we arrive at the expression for the original function $f(x)$ by integrating $f^{(n)}(x)$ a total of n times:

$$f(x) = \int^n f^{(n)}(x) dx = e^x \int e^x \int \cdots \int x^x e^{-x} Y_n(x) dx \quad (\text{n nested integrals})$$

This provides a recursive, structured method of reverse-engineering the integral of x^x using its known derivative structure, and reveals the compositional depth of the function through repeated integration of Bell-polynomial-weighted terms.

Operator Approach

In the previous section, we derived a recursive formula for the n th derivative of $f(x) = x^x$ using Bell polynomials. We now revisit the reverse-engineering process in a more compact and elegant way using operator notation.

Rewriting the Derivative Structure with Operators. We observed earlier that:

$$f^{(n)}(x) = x^x \cdot Y_n(x) \quad \text{and} \quad f^{(n-1)}(x) = x^x \cdot A_{n-1}(x),$$

where the coefficient function $A_{n-1}(x)$ satisfies the differential equation:

$$A'_{n-1}(x) + (\ln x + 1) A_{n-1}(x) = Y_n(x).$$

Define the differential operator:

$$L := \frac{d}{dx} + (\ln x + 1).$$

Then the equation becomes:

$$L A_{n-1}(x) = Y_n(x), \quad \Rightarrow \quad A_{n-1}(x) = L^{-1} Y_n(x).$$

Inverse Operator via Integrating Factor. We previously computed the integrating factor:

$$\mu(x) = e^{\int (\ln x + 1) dx} = x^x e^{-x}.$$

Using this, the inverse operator L^{-1} can be explicitly expressed as:

$$L^{-1} Y_n(x) = \frac{1}{x^x e^{-x}} \int x^x e^{-x} Y_n(x) dx.$$

Therefore, the $(n-1)$ st derivative of $f(x)$ can be compactly written as:

$$f^{(n-1)}(x) = x^x \cdot L^{-1} Y_n(x).$$

Recursive Integration via Operator Powers. Continuing this logic, the original function $f(x)$ can be obtained from $f^{(n)}(x)$ by applying the inverse operator L^{-1} a total of n times:

$$f(x) = (L^{-1})^n [x^x \cdot Y_n(x)].$$

This formulation neatly encodes the full structure of the recursive integration in a symbolic operator form, avoiding the explicit writing of n nested integrals.

Comment on Generating Functions (Optional Extension). As a further abstraction, we can define a generating function for the derivatives:

$$G(t, x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x)}{n!} t^n = x^x \sum_{n=0}^{\infty} \frac{Y_n(g'(x), g''(x), \dots)}{n!} t^n.$$

This viewpoint connects the entire family of derivatives with a single formal power series, offering potential for closed-form manipulation or approximation.

This operator approach not only streamlines the integration process but also reveals the algebraic structure underlying the recursive relationship between successive derivatives of x^x .

First Integral of $f(x) = x^x$

We now apply the operator-based reverse engineering technique to compute the first integral of x^x , i.e., to find a function $F(x)$ such that:

$$F'(x) = x^x$$

Recall the operator:

$$L := \frac{d}{dx} + (\ln x + 1), \quad \text{with inverse: } L^{-1}[f](x) = \frac{1}{x^x e^{-x}} \int x^x e^{-x} f(x) dx$$

Let $A(x) = F(x)/x^x$. Then:

$$F(x) = x^x A(x) \quad \Rightarrow \quad F'(x) = x^x (A'(x) + (\ln x + 1)A(x))$$

Comparing with $F'(x) = x^x$, we obtain:

$$A'(x) + (\ln x + 1)A(x) = 1 \quad \Rightarrow \quad LA(x) = 1 \quad \Rightarrow \quad A(x) = L^{-1}[1](x)$$

Using the integrating factor $\mu(x) = x^x e^{-x}$, we compute:

$$A(x) = \frac{1}{x^x e^{-x}} \int x^x e^{-x} \cdot 1 dx = \frac{1}{x^x e^{-x}} \int x^x e^{-x} dx$$

Thus, the first integral is:

$$\int x^x dx = F(x) = x^x \cdot \frac{1}{x^x e^{-x}} \int x^x e^{-x} dx = e^x \int x^x e^{-x} dx$$

Final Result:

$$\boxed{\int x^x dx = e^x \int x^x e^{-x} dx + C}$$

This expression provides a clean representation of the integral of x^x in terms of itself weighted by an exponential kernel. While not elementary, it reveals a structural insight useful for approximations, asymptotic expansions, or recursive integration.

Antiderivative of x^x Without Explicit Integration

We now aim to express the integral of x^x without writing it in terms of an explicit integral sign. To achieve this, we consider the differential operator:

$$L := \frac{d}{dx} + (\ln x + 1)$$

This operator appeared naturally in the recursive structure of the derivatives of $f(x) = x^x$, and its inverse provides a direct route to the antiderivative.

We seek a function $F(x)$ such that:

$$L[F(x)] = x^x \quad \Rightarrow \quad F'(x) + (\ln x + 1)F(x) = x^x$$

This is a first-order linear ODE, and its solution gives the antiderivative of x^x . Thus, we define:

$$F(x) := L^{-1}[x^x]$$

That is:

$$\boxed{\int x^x dx = (L^{-1})[x^x]} \quad \text{where } L = \frac{d}{dx} + (\ln x + 1)$$

This compact expression eliminates the explicit integral symbol and encodes the structure of the antiderivative entirely in terms of the inverse differential operator.

This approach aligns with the recursive and operator-based methods used for the higher-order derivatives, and gives a formally elegant representation of the indefinite integral of x^x .

Evaluating $\int_0^1 x^x dx$ via Operator Method

Goal. We aim to evaluate the definite integral:

$$\int_0^1 x^x dx$$

Step 1: Use the Antiderivative Formula. From previous work using operator methods, we derived the antiderivative:

$$\int x^x dx = e^x \int x^x e^{-x} dx + C$$

Applying this to the definite integral:

$$\int_0^1 x^x dx = \left[e^x \int x^x e^{-x} dx \right]_0^1$$

Let:

$$I := \int_0^1 x^x dx \quad \text{and} \quad J := \int_0^1 x^x e^{-x} dx$$

Then:

$$I = e \cdot J$$

Step 2: Compute $J = \int_0^1 x^x e^{-x} dx$. This integral has no elementary closed-form, but it can be computed numerically:

$$J \approx 0.47503568433$$

Step 3: Multiply by e . Using the approximation $e \approx 2.71828$, we find:

$$I = e \cdot J \approx 2.71828 \cdot 0.47503568433 \approx 1.29128$$

Final Result.

$$\boxed{\int_0^1 x^x dx = e \cdot \int_0^1 x^x e^{-x} dx \approx 1.29128}$$

Summary. Using the identity:

$$\int x^x dx = e^x \int x^x e^{-x} dx + C$$

we expressed the definite integral as a product of e and a non-elementary weighted exponential integral, then evaluated it numerically. This approach highlights the power of operator techniques in symbolic integration.

2. Concluding Remarks

Although the function x^x is not elementary in the sense of classical integrability, its derivative structure reveals a rich interplay between exponential and logarithmic behaviors. The operator method presented here is both elegant and versatile, offering insights into other non-elementary functions of similar form.

The techniques employed—from Bell polynomials to integrating factors—are standard, yet their synthesis in this context may be instructive. Such methods may also be extended to functions of the form $x^{g(x)}$ or $g(x)^x$, inviting further exploration.